INCH-POUND
MIL-STD-798A(SH)
23 January 2019
SUPERSEDING
MIL-STD-798(SHIPS)
16 December 1965

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

DESIGN EVALUATION REQUIREMENTS FOR VALVES FOR NAVAL SHIPBOARD USE



AMSC N/A FSC 4820

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. This standard establishes the requirements for nondestructive testing (NDT), welding, material control and identification, vibration, and high-impact (H.I.) shock test requirements for valves for naval shipboard use.

3. This standard may be used as an advisory document regarding valve design and testing when not specifically referenced.

4. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH PAGE
1. SCOPE 1
1.1 Scope 1
1.2 Applicability 1
2. APPLICABLE DOCUMENTS 1
2.1 General 1
2.2 Government documents 1
2.2.1 Specifications, standards, and handbooks 1
2.2.2 Other Government documents, drawings, and publications 1
2.3 Non-Government publications 2
2.4 Order of precedence 2
3. DEFINITIONS 2
3.1 Actual seating surface 2
3.2 Body 2
3.3 Cast 2
3.4 Cold-rated working pressure 2
3.5 Ferrous component primary service rating 2
3.6 Finished surface 2
3.7 Forged 2
3.8 Included seating surface 2
3.9 Indication 2
3.10 Linear indications 2
3.11 Manually operated valve 2
3.12 Nonferrous component primary service rating 3
3.13 Nominal pipe size (nps) 3
3.14 Pressure containing sealing surface 3
3.15 Principal unit 3
3.16 Size 3
3.17 Technical Authority 3
3.18 Valve actuator 3
3.19 Valve and actuator principal unit (VAPU) 3
3.20 Valve and operator principal unit (VOPU) 3
3.21 Valve bonnet 3
3.22 Valve operator 3
3.23 Valve principal unit (VPU) 3
3.24 Valve shock acceptance 3
3.25 Valve shock qualification 3
3.26 Valve subsidiary component (VSC) 3
3.27 Wall thickness 3
4. GENERAL REQUIREMENTS 3
4.1 Design evaluation or FAT 3
4.2 Recycled, recovered, environmentally preferable, or biobased materials 3
4.3 Welding and examinations of forgings 4
5. DETAILED REQUIREMENTS 4
5.1 Radiography 4
5.2 Magnetic particle inspection 4
5.3 Liquid penetrant inspection 5
5.4 Welding 5
5.4.1 Welding execution 5
5.4.2 Nondestructive testing (NDT) of welds 5
5.4.3 Radiographic acceptance 5
5.4.4 Liquid penetrant and magnetic particle acceptance 5
5.5 Material control 6
5.5.1 Assembled valve material control 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

5.5.2 Separate valve parts material control 6
5.6 Material identification 6
5.6.1 Visibility of markings 6
5.6.2 Post-manufacture marking 6
5.6.3 Tools and method for marking 6
5.7 H.I. shock tests 7
5.7.1 Principal unit H.I. shock tests 7
5.7.2 Sizes to be tested 7
5.7.3 Pre-shock examination and testing 7
5.7.4 Shock test operating positions and conditions 7
5.7.5 Adjustments during shock tests 8
5.7.6 Post-shock test examination and testing 8
5.7.7 Shock acceptance criteria 9
5.7.8 Shock test extension 9
5.7.8.1 Identical shock test extension 9
5.7.8.2 Similar shock test extension 9
5.7.8.3 Additional extension criteria for VAPUs and VOPUs 9
5.7.8.4 Shock test extension guidance for VSCs 9
5.8 Vibration test 10
5.8.1 Valves and valve actuators to be tested 10
5.8.2 Pre-vibration examination and testing 10
5.8.3 Conduct of vibration test 10
5.8.4 Vibration test operating positions and conditions 10
5.8.5 Post-vibration test examination and acceptance 10
5.8.6 Extension of vibration test 11
5.9 Post-shock and post-vibration test 11
6. NOTES 11
6.1 Intended use 11
6.2 Acquisition requirements 11
6.3 Subject term (key word) listing 11
6.4 Changes from previous issue 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard covers the design evaluation or first article testing (FAT) requirements for new families of valves; modifications to existing valves, including nondestructive, welding, material identification, and vibration; and high-impact (H.I.) shock testing for valves and their actuators. Design evaluation or FAT is performed on a valve, multiple valves, or a series of valves in the same family to ensure the design will meet requirements for the intended ship systems. It does not apply to primary side components of nuclear ships.

1.2 Applicability. This standard applies to valves of all service ratings.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0924-LP-062-0010	-	Submarine Safety (SUBSAFE) Requirements Manual
0948-LP-045-7010	-	Material Control Standard, Volume 1
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E186	-	Standard Reference Radiographs for Heavy-Walled (2 to 4½ in. (50.8 to 114 mm)) Steel Castings
ASTM E272	-	Standard Reference Radiographs for High-Strength Copper-Base and Nickel-Copper Alloy Castings
ASTM E280	-	Standard Reference Radiographs for Heavy-Walled (4½ to 12 in. (114 to 305 mm)) Steel Castings
ASTM E310	-	Standard Reference Radiographs for Tin Bronze Castings
ASTM E446	-	Standard Reference Radiographs for Steel Castings Up to 2 in. (50.8 mm) in Thickness

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

(This section contains a number of definitions that are provided for clarity in the valves design field. These terms may not be present elsewhere in this standard.)

3.1 Actual seating surface. The portion of a valve, trap seat, or seating face that comes in direct contact with a mating face, including ⅛ inch on either side of the direct contact area.

3.2 Body. The main pressure enclosure, including bonnet and cover attachment area.

3.3 Cast. An object made by shaping molten metal or similar material in a mold.

3.4 Cold-rated working pressure. The maximum operating pressure as defined by the pressure class rating at 100 °F.

3.5 Ferrous component primary service rating. The hot pressure rating of a ferrous component.

3.6 Finished surface. The surfaces that are finished machined in their entirety.

3.7 Forged. Items for which the manufacturing process includes shaping metal using localized compressive forces.

3.8 Included seating surface. The portion of a valve, trap seat, or seating face that borders the actual seating face but does not come into contact with the mating seat or face. This surface is finished machined in the same manner as the actual seating face.

3.9 Indication. An interruption of the typical structure of a material, such as a lack of homogeneity in its mechanical, metallurgical, or physical characteristics.

3.10 Linear indications. Defects that have a length-to-width ratio equal to or exceeding three, where the length equals ⅛ inch or more, and that cannot be defined as porosity or shrink.

3.11 Manually operated valve. A valve operated by hand using a handwheel, lever, or similar component that directly (i.e., without additional gearing) causes stem motion. A gearbox-operated valve, though hand operated, is not considered a manually operated valve herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

3.12 Nonferrous component primary service rating. The cold pressure rating of a nonferrous component.

3.13 Nominal pipe size (nps). The set of standard sizes used for pipe, fittings, and valves.

3.14 Pressure containing sealing surface. All of the surfaces that come into direct contact with the sealing member.

3.15 Principal unit. For the purpose of shock testing and shock extension, items that are directly supported by the ship structure or by a foundation that is directly attached to the ship structure, and items mounted in piping systems, ducting systems, and similar systems that are supported by ship structure.

3.16 Size. The component nps.

3.17 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. Contact NAVSEA 05S for assistance with contacting the appropriate Technical Authority for U.S. Navy ships.

3.18 Valve actuator. An electric, hydraulic, mechanical (gearbox), or pneumatic means of remotely operating a valve. A subsidiary component of a VAPU (see 3.19).

3.19 Valve and actuator principal unit (VAPU). For the purpose of shock testing and shock extension, a valve subsidiary component combined with an integral valve actuator subsidiary component (see 3.18).

3.20 Valve and operator principal unit (VOPU). For the purpose of shock testing and shock extension, a valve subsidiary component combined with an integral valve operator subsidiary component (see 3.22).

3.21 Valve bonnet. All pressure containing sections from the body attachment area up to, but not including, the packing gland area.

3.22 Valve operator. A local means of operating a valve, including handwheels and levers. A subsidiary component of a VOPU (see 3.20).

3.23 Valve principal unit (VPU). For the purpose of shock testing and shock extension, a valve assembly without a valve actuator or valve operator (see 3.18 and 3.22).

3.24 Valve shock acceptance. The determination that a VPU, VOPU, VAPU, or valve subsidiary component (VSC) has demonstrated compliance with shock acceptance criteria of applicable generic acquisition documents (e.g., FAT, military specifications, general procurement specifications). Shock acceptance via documentation is not the same as shock qualification.

3.25 Valve shock qualification. The determination that a VPU, VOPU, or VAPU has demonstrated compliance with shock acceptance criteria for a particular service and installation on a particular ship hull (or ship class).

3.26 Valve subsidiary component (VSC). For the purpose of shock testing and shock extension, the valve in a VAPU or VOPU (see 3.19 or 3.20).

3.27 Wall thickness. The section thickness under examination or otherwise being discussed or worked.

4. GENERAL REQUIREMENTS

4.1 Design evaluation or FAT. This standard shall be used in the design and construction of valves intended for use on naval vessels. All requirements herein shall be met by new designs and during the FAT process for any new product valves.

4.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

4.3 Welding and examinations of forgings. Where welding and examination procedures are not specifically delineated in this standard or in referenced documents for forgings, those procedures shall follow the same requirements as those stated for castings.

5. DETAILED REQUIREMENTS

5.1 Radiography. Radiographic examination shall be performed in accordance with T9074-AS-GIB-010/271. It shall be applied to all castings for bodies, bonnets, covers, tailpieces, and discs for quick opening valves or relief valves covered in this standard when both the size and primary pressure rating of the component equal or exceed the values listed in table I. The acceptance criteria for this examination shall be in accordance with table II. Acceptance class criteria shall be in accordance with MIL-STD-2035.

TABLE I. Size and pressure rating.

Item	Material	Size	Primary Pressure Rating
Valves, flanges, fittings, and auxiliary equipment (except for hydrogen and oxygen services)	Ferrous and nonferrous	Over 2 inches nps	300 psi 1/ and higher
		½ inch nps and larger	Over 3,000 psi
Valves, flanges, fittings, and auxiliary equipment (for hydrogen and oxygen services)	Ferrous and nonferrous	½ inch nps and larger	550 psi and higher

NOTE:
1/ psi = pounds per square inch.

TABLE II. Acceptance criteria.

Alloy Type	Wall Thickness	Acceptance Standard	Acceptance Class per MIL-STD-2035
Ferrous	Less than 1 inch	ASTM E446	Class 1
	1 to 2 inches		Class 2
	2 to 4½ inches	ASTM E186	Class 2
	4½ to 12 inches	ASTM E280	Class 2
Nickel base, copper, or aluminum bronze	Less than 1 inch	ASTM E272	Class 2
	1 inch and over		Class 3
Tin bronze	½ inch and less	ASTM E310	Class 2 with no area restriction or shrinkage
	Over ½ inch	ASTM E310	Class 3, except class 4 shrinkage

5.2 Magnetic particle inspection. Magnetic particle inspection shall be performed in accordance with T9074-AS-GIB-010/271 on ferrous components only. Inspection shall be performed in accordance with table III. Acceptance criteria shall be in accordance with table III. Inspection shall be applied when both the size and primary pressure rating exceed the values shown and shall be performed after rough machining and final heat treatment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

TABLE III. Magnetic particle inspection and acceptance criteria.

Item	Size	Primary Pressure Rating	Material	Area to be Inspected	Acceptance Criteria
All items	½ inch and larger	225 psi and higher	Cast, forged	Butt weld ends	Remove all indications
Valves, auxiliary equipment (bodies only)	½ inch and larger (see acceptance criteria)	600 psi and higher	Cast, forged	Pressure containing sealing surface (see acceptance criteria)	2 inches and smaller: remove all linear indications
					Over 2 inches; external and accessible internal surface: remove all linear indications

5.3 Liquid penetrant inspection. Liquid penetrant inspection shall be performed in accordance with T9074-AS-GIB-010/271. Inspection shall be performed in accordance with table IV when both the size and pressure rating exceed the values shown and after final heat treatment. Acceptance criteria shall be in accordance with table IV.

TABLE IV. Liquid penetrant inspection and acceptance criteria.

Item	Location (Surface) to be Inspected	Size	Primary Pressure Rating	Material	Surface Condition	Acceptance Criteria
All	Butt weld end preparations	½ inch and larger	225 psi and higher	Nonferrous	Finished machined	Remove all indications
All	Actual seating surfaces	All	All	All	Finished machined	Remove all indications
All	Included seating surfaces	All	All	All	Finished machined	Remove all linear indications
Weld inlayed or overlayed surfaces	Pressure containing sealing surface	All	All	All	Finished machined	Remove all indications

5.4 Welding. Welding conducted for all items under this standard shall comply with all the requirements of S9074-AR-GIB-010/278 for both production and repair welding on castings and forgings.

5.4.1 Welding execution. Welding shall not be performed until the applicable procedures have been approved and the applicable welders have obtained the necessary qualification approvals.

5.4.2 Nondestructive testing (NDT) of welds. Radiographic, magnetic particle, and liquid penetrant NDT inspection of welds as required shall be performed in accordance with T9074-AS-GIB-010/271.

5.4.3 Radiographic acceptance. The radiographic examination acceptance criteria shall be in accordance with MIL-STD-2035.

5.4.4 Liquid penetrant and magnetic particle acceptance. The liquid penetrant and magnetic particle inspection acceptance criteria of welds shall be in accordance with MIL-STD-2035, class 1, except as otherwise noted herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

5.5 Material control. When specified (see 6.2), all materials under this standard shall be in accordance with 0948-LP-045-7010 and Level 1 controls, and, when specified, 0924-LP-062-0010.

5.5.1 Assembled valve material control. Material control shall be required for the following parts when valves are shipped in the assembled form:

a. Valve bodies.

b. Valve bonnets.

c. Valve tailpieces (including tailpiece attachments).

d. Valve bonnet fasteners (nuts, bolts, union nuts).

e. End connection union nuts.

f. Valve covers (access plates and other removable covers).

g. Valve flanges.

h. Valve fittings.

i. Valve stems.

5.5.2 Separate valve parts material control. Material control shall be required for the following valve repair parts when the repair parts are furnished separately:

a. Balls, discs, or wedges.

b. Springs.

c. Gaskets, O-ring seals, and seal rings.

d. Valve seats.

e. Stem bushings and washers.

f. Seal retainers.

g. Parts listed in 5.5.1.

5.6 Material identification. If not specified in material ordering specifications, materials, items, and parts shall be marked and identified as specified in this section. Material identification marks or markings shall be required for all valve assemblies, the designated valve repair parts, and for any item designated as “Level 1”, as specified (see 6.2).

5.6.1 Visibility of markings. All material identification marks applied to the parts listed in 5.5.1, except for valve stems, shall be visible after assembly.

5.6.2 Post-manufacture marking. All marking, when not applied at the time of manufacture, such as a casting heat number, shall be placed on a raised pad or similar low stress area and shall be permanent.

5.6.3 Tools and method for marking. The following shall apply to marking tools and methods:

a. Permanent marking shall be applied by a method that will not reduce thickness below minimum design requirements, induce detrimental stresses, contaminate, or cause maloperation.

b. Marking with impact tools is permitted only on raised pads or low stress areas and only when round nosed tools are used. The impression shall not exceed 0.010 inch.

c. Electric arc etching is prohibited.

d. Vibrating pencil or electro-chemical etching are permitted on items and assemblies that cannot be impact marked due to size or shape.

e. Roll marking is permitted on bar stock where the die imprint shall not occur on a critically stressed surface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

5.7 H.I. shock tests. Principal units shall be tested in accordance with MIL-DTL-901 and as follows:

5.7.1 Principal unit H.I. shock tests. VPUs, VAPUs, and VOPUs shall be H.I. shock tested with operating positions and conditions as specified in table V. The VPU or VSC shall be subjected to a pressure (the test media shall be at ambient or room temperature) equal to the cold-rated working pressure of the item. VAPUs and VOPUs shall be shock tested with their actuator or operator, unless otherwise approved by the Technical Authority (see 3.17). VAPUs and VOPUs shock tested with a dummy mass in lieu of their respective actuator or operator shall meet the shock test extension requirements of MIL-DTL-901 (see 5.7.8).

5.7.2 Sizes to be tested. Valve test sizes conform to the following nps in inches. When testing multiple sizes of the same design, only the following sizes need to be tested, except that the largest size must also be tested (see 5.7.8.4). These sizes are used when no specific test sizes are called out in applicable valve specifications.

a. ¾ inch.

b. 2 inches.

c. 4 inches.

d. 8 inches.

e. 12 inches.

f. All sizes larger than 12 inches.

5.7.3 Pre-shock examination and testing. Pre-shock test examination and operational tests are required in all cases. The tested unit shall receive a complete visual and dimensional examination, a functional check, and a hydrostatic shell and seat tightness test before the shock test series as specified in 5.7.3.a through 5.7.3.d. The tests shall be conducted in the order as follows:

a. An external visual and dimensional check without disassembly shall be conducted prior to the first blow, shot, or drop of the shock test series.

b. A functional check consisting of cycling the valve three complete cycles. A complete cycle is defined as going from one position to the other and back.

c. A hydrostatic shell test to 150 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted prior to the first blow, shot, or drop of the shock test series. This pressure shall be maintained for a minimum of 10 minutes.

d. A seat tightness test to 110 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted prior to the first blow, shot, or drop of the shock test series. This pressure shall be maintained for at least 3 minutes.

5.7.4 Shock test operating positions and conditions. Valve positions and conditions shall be as specified in table V during H.I. shock testing. Operating modes shall be defined in the applicable shock test procedure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

TABLE V. Operating positions and conditions of VPUs or VSC.

Valve Subsidiary Component Type	Positions	Pressure Conditions
Non-throttling (gate, ball, butterfly, trip, solenoid, etc.)	Full open	Pressurized throughout
	Full closed	Pressurized on inlet side
Throttling (globe, angle, dump, astern, needle, sleeve, temperature regulating, etc.)	Full closed	Pressurized on inlet side
	Disc at intermediate position (not on either seat)	Pressurized throughout
Check (swing lift)	As-is	Pressurized throughout
	Full closed	Back pressure on disc
Stop check	As-is, stem backed off	Pressurized throughout
	Full closed, stem engaged	Pressurized on inlet side
	Full closed, stem backed off	Back pressure on disc
Automatic pressure regulating and control 1/	Lock up	Pressurized on inlet side, valve maintaining outlet pressure in a dead-end volume, equal to mid-range setting of the valve
Relief 2/	Full closed	Pressurized on inlet side

NOTES:

1/ The validity of shock test approval of pressure regulating or control VPUs or VSCs tested at the "maximum design pressure differential" as required by the previous issue of this standard is not affected by this change.

2/ Relief, sentinel, and safety VPUs or VSCs shall be tested in the full closed position only. The VPU or VSC shall be tested at the upper setting of the heaviest spring and the lowest setting of the lightest spring for which the VPU or VSC is designed. VSCs designed for only one spring shall be tested at the highest and lowest spring setting. The hydrostatic pressure during shock testing shall be equal to the reseat pressure specified for the spring setting being tested. This test may be used to support shock extension of VPUs or VSCs for the entire series of springs and spring settings for that size and design.

5.7.5 Adjustments during shock tests. Adjustments to correct minor malfunctions during shock tests will be permitted within the scope of what a person on watch could be expected to do without leaving their station (for example, tightening a packing gland is permissible but replacing a yielded or sheared bolt is not). Pass/fail criteria shall be met prior to any correction of minor malfunctions.

5.7.6 Post-shock test examination and testing. Post-shock test examination and operational tests are required in all cases. The tested unit shall receive a complete visual and dimensional examination after the shock test series as specified in 5.7.6.a through 5.7.6.d. The tests shall be conducted in the order as follows:

a. An external visual and dimensional check without disassembly. When specified (see 6.2), a liquid penetrant or magnetic particle inspection shall be conducted at the completion of the shock test series.

b. A functional check consisting of cycling the valve three complete cycles. A complete cycle is defined as going from one position to the other and back.

c. A hydrostatic shell test to 150 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted at the completion of the shock test series. This pressure shall be maintained for a minimum of 10 minutes.

d. A seat tightness test to 110 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted at the completion of the shock test series. This pressure shall be maintained for at least 3 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

5.7.7 Shock acceptance criteria. The tested VPU, VOPU, or VAPU shall pass all post-shock tests without exception. During the visual and dimensional checks, any permanent damage detrimental to the principal unit meeting its pass/fail criteria shall be cause for rejection. This shall include any permanent deformation of any pressure containing part. Shock acceptance criteria may be generic or applicable to a particular ship acquisition program. In cases where valves are procured to general specifications, a shock extension shall be required to demonstrate that it complies with an applicable ship's specifications and that it meets the requirements for shock test extension stated in MIL-DTL-901.

5.7.8 Shock test extension. Shock test extension is applicable to VPUs, VOPUs, or VAPUs procured to general procurement specifications (e.g., military specifications) or to principal units procured for a particular ship (or ship class) acquisition program. Shock test extensions shall comply with the requirements of MIL-DTL-901. The conditions in table VI shall be met to permit shock test extension of an untested principal unit from a shock tested VPU with a VPU or VSC of a different size. In cases where general procurement specifications (e.g., a valve military specification) state extension criteria, those criteria shall supersede the extension criteria of this standard.

5.7.8.1 Identical shock test extension. The following criteria are applicable to identical extensions procured to a general procurement specification or procured for a particular ship hull (or ship class):

a. Same geometry and design.
b. Same materials.
c. Same pressure rating/service.
d. Same bonnet construction.
e. Same end connection construction.
f. Same methods of sealing throughout.
g. Same mode of operation, operator, and actuator attachment.
h. Same manufacturing processes.
i. Same installation/mounting location aboard ship and service/mission as represented in the shock test.

(NOTE: 5.7.8.1.i is not applicable to principal units procured to a general procurement specification where no shipboard installation, mounting, service, or mission is identified in the contract documents.)

5.7.8.2 Similar shock test extension. When specified (see 6.2), or when not all conditions of 5.7.8 are met, shock test extensions shall meet the shock test extension requirements of MIL-DTL-901, demonstrating that the untested item has equal or greater shock resistance than the tested item with respect to the criteria outlined in 5.7.8.1.a through i. 5.7.8.1.i is not applicable to VPUs procured to a general procurement specification where no shipboard installation, mounting, service, or mission is identified in the contract documents.

5.7.8.3 Additional extension criteria for VAPUs and VOPUs. A shock-tested VAPU may be used as a basis for shock extension of a VPU, or VOPU with a VSC, of the same size or smaller design (see 5.7.8.4). VOPUs shall not be used as a basis for shock test extension of VAPUs but may be used as a basis for shock extension of a VPU. VAPUs shall not be extended from VAPUs with a different actuator type. For example, extension of a VAPU with a hydraulic actuator from a VAPU with an electric actuator is not permitted. In cases where shock test extension candidate and basis VAPUs have different actuator types, subsidiary component/subassembly testing of the candidate actuator may be performed to support shock test extension in accordance with MIL-DTL-901. If shock test extension requirements of MIL-DTL-901 cannot be met, the VAPU shall be tested.

5.7.8.4 Shock test extension guidance for VSCs. Table VI provides a representative listing of VPU or VSC sizes that, when shock tested, may be used as a basis for shock test extension. Sizes not listed in table VI may have the requirements of this section applied when specified (see 6.2). Table VI shock test extension candidate size ranges are based on the general situation throughout the valve industry where the VPU or VSC design is proportionally similar to the design tested; VPU and VSC sizes are discretized to sizes where significant change occurs in the design. VPU or VSC design series at sizes other than those listed in table VI may be used to support shock extension with Technical Authority (see 3.17) approval.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

TABLE VI. Test sizes and applicable extension sizes.

Size Tested (Inches)	Shock Extension Candidates (Inches) (see 5.7.1)
¾	Smaller than 1
2	1 to 2, inclusive
4	2½ to 4, inclusive
8	6, 7, and 8
12	9, 10, and 12
Over 12	All sizes shall be tested.
NOTE: Sizes 7 and 9 are not normally used by the United States Navy.	

5.8 Vibration test. Valves and valve actuators shall be tested in accordance with MIL-STD-167-1 and as follows.

5.8.1 Valves and valve actuators to be tested. Valves, with actuators installed if applicable, of the sizes listed in 5.7.2 shall be tested. Sizes other than those tested shall be approved as specified in table VI. Sizes other than those tested shall be subject to Technical Authority (see 3.17) approval using the same criteria for other sizes as specified for shock extensions in table VI.

5.8.2 Pre-vibration examination and testing. Pre-vibration test examination and operational tests are required in all cases. The tested unit shall receive a complete visual and dimensional examination, a functional check, and a hydrostatic shell and seat tightness test before the vibration test series as specified in 5.8.2.a through 5.8.2.d. The tests shall be conducted in the following order:

a. An external visual and dimensional check without disassembly prior to the vibration test series.

b. A functional check consisting of cycling the valve three complete cycles. A complete cycle is defined as going from one position to the other and back.

c. A hydrostatic shell test to 150 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted prior to the vibration test series. This pressure shall be maintained for a minimum of 10 minutes.

d. A seat tightness test to 110 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted prior to the vibration test series. This pressure shall be maintained for at least 3 minutes.

5.8.3 Conduct of vibration test. Valves, with actuators installed if applicable, shall meet the type I requirements of MIL-STD-167-1 under the applicable conditions of table V.

5.8.4 Vibration test operating positions and conditions. Valve positions and conditions shall be as specified in table V during vibration testing. Operating modes shall be defined in the applicable vibration test procedure.

5.8.5 Post-vibration test examination and acceptance. Post-vibration test examination and operational tests are required in all cases. The tested unit shall receive a complete visual and dimensional examination after the vibration test series as specified in 5.8.5.a through 5.8.5.d. The tests shall be conducted in the following order:

a. An external visual and dimensional check without disassembly at the completion of the vibration test series.

b. A functional check consisting of cycling the valve three complete cycles. A complete cycle is defined as going from one position to the other and back.

c. A hydrostatic shell test to 150 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted at the completion of the vibration test series. This pressure shall be maintained for a minimum of 10 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.

d. A seat tightness test to 110 percent of the full cold-rated working pressure, or as defined in the valve specification, shall be conducted at the completion of the vibration test series. This pressure shall be maintained for at least 3 minutes.

5.8.6 Extension of vibration test. Vibration testing extension may be permitted in accordance with MIL-STD-167-1 or NAVSEA Technical Authority (see 3.17) approval. Tests shall be conducted with the smallest valve/largest actuator combination assembly being qualified. Successful test results shall be extended to combinations of the same actuator mounted on larger valves.

5.9 Post-shock and post-vibration test. Upon completion of all shock and vibration testing, a final post-test examination shall be conducted. In addition to the functional check, shell test, and seat tightness tests, a final visual and dimensional check including disassembly and examination of valve internals shall be conducted at the completion of the shock and vibration test series.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The information covered by this standard is intended for use in the design and construction of valves. Naval vessel construction and the demands of service differ from commercial design practices. This standard accounts for the potential hazards encountered and consideration for battle damage and mission essential system reliability.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Material controls required (see 5.5).

c. Material identification required (see 5.6).

d. Shock testing sizes, if not specified herein (see 5.7.2).

e. Requirement for post-shock testing liquid penetrant or magnetic particle inspection (see 5.7.6.a).

f. Additional shock test requirements (see 5.7.8.2).

g. Vibration testing sizes, if not specified herein (see 5.8.1).

6.3 Subject term (key word) listing.

Extension

Liquid penetrant

Magnetic particle

Radiographic

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4820-2018-029)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:44Z
Check the source to verify that this is the current version before use.